

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12387517
B1
August 12, 2025
Kittenplon; Yair et al.

Graph-based document layout detection

Abstract

A document layout system for determining a layout of a document. The document layout system is configured to apply an OCR technique to identify the text in the document. The document layout system is further configured to generate a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text. The document layout system is also configured to apply a graph cluster network machine learning model to the graph representation to identify a layout of different sections of the document according to respective merge inferences determined for individual ones of the plurality of edges. The document layout system is also configured to provide the layout of different sections of the document.

Inventors: Kittenplon; Yair (Raanana, IL), Mazor; Shai (Binyamina, IL), Litman; Ron (Tel-Aviv, IL), Nayman; Niv Berchi (Shoham, IL), Tsiper; Shahar (Haifa, IL)

Applicant: Amazon Technologies, Inc. (Seattle, WA)

Family ID: 96662004

Assignee: Amazon Technologies, Inc. (Seattle, WA)

Appl. No.: 18/065541

Filed: December 13, 2022

Publication Classification

Int. Cl.: G06V30/414 (20220101); G06V10/82 (20220101); G06V30/10 (20220101);
G06V30/413 (20220101); G06V30/416 (20220101)

U.S. Cl.:

CPC **G06V30/414** (20220101); **G06V10/82** (20220101); **G06V30/10** (20220101);
G06V30/413 (20220101); **G06V30/416** (20220101);

Field of Classification Search

CPC: G06V (10/82); G06V (30/10); G06V (30/41); G06V (30/412); G06V (30/413); G06V (30/414); G06V (30/416); G06F (16/93); G06F (16/9024)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8565474	12/2012	Radakovic	N/A	N/A
10049270	12/2017	Agarwalla	N/A	G06V 30/414
11087086	12/2020	An	N/A	G06F 18/24
11113518	12/2020	Chua	N/A	G06F 18/2413
2013/0194448	12/2012	Baheti	N/A	N/A
2023/0146501	12/2022	Agarwal	382/190	G06N 3/09
2023/0260310	12/2022	Rangarajan	382/176	G06V 30/414
2023/0394859	12/2022	Montero	N/A	G06V 30/146

OTHER PUBLICATIONS

Renshen Wang et al., “Post-OCR Paragraph Recognition by Graph Convolution Networks”, dated Nov. 15, 2022, pp. 1-11. cited by applicant

Primary Examiner: Mariam; Daniel G

Attorney, Agent or Firm: Kowert, Hood, Munyon, Rankin & Goetzel, P.C.

Background/Summary

BACKGROUND

(1) Documents may be stored in a digital format on computing systems. The documents may also be converted from a physical document to a digital document based on optically capturing the physical document, such as by scanning the physical document. Scanning software may also include optical character recognition (OCR) functionality that allows the digital document to be processed or interpreted as a text-based document by the computing systems.

(2) Text-based documents do not conventionally include layout information that identify a natural reading order that is perceptible by a human reader. OCR algorithms may be unable to discern that a document that has a two-column layout should be read based on the columns rather than as lines going across the entire page. Additionally, the OCR algorithms may be unable to determine that the document is for a particular type of document based on its layout.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of system configured to determine text section categories and layouts of documents, according to some embodiments.
- (2) FIG. 2 illustrates representations of a document being processed by a document layout system, according to some embodiments.
- (3) FIG. 3 illustrates graph representations of documents processed by one or more aspects of a document layout system, according to some embodiments.
- (4) FIG. 4 illustrates a block diagram of bounding boxes and that represent text elements in a document that is processed by a document layout system, according to some embodiments.
- (5) FIG. 5 is a logical block diagram illustrating a provider network that implements different services including a code development service, according to some embodiments.
- (6) FIG. 6 illustrates an interface for a document layout service, according to some embodiments.
- (7) FIG. 7 is a block diagram of a method for determining a document layout based on text elements contained within the document, according to some embodiments.
- (8) FIG. 8 is a block diagram showing an example computing device to implement the various techniques described herein, according to some embodiments.
- (9) While embodiments are described herein by way of example for several embodiments and illustrative drawings, those skilled in the art will recognize that the embodiments are not limited to the embodiments or drawings described. It should be understood that the drawings and detailed description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include”, “including”, and “includes” mean including, but not limited to.
- (10) “Based On.” As used herein, this term is used to describe one or more factors that affect a determination. This term does not foreclose additional factors that may affect a determination. That is, a determination may be solely based on those factors or based, at least in part, on those factors. Consider the phrase “determine A based on B.” While B may be a factor that affects the determination of A, such a phrase does not foreclose the determination of A from also being based on C. In other instances, A may be determined based solely on B.
- (11) This specification includes references to “one embodiment” or “an embodiment.” The appearances of the phrases “in one embodiment” or “in an embodiment” do not necessarily refer to the same embodiment. Particular features, structures, or characteristics may be combined in any suitable manner consistent with this disclosure.

DETAILED DESCRIPTION

- (12) Various systems and methods for detecting layouts for documents based on graph representations of the documents are described herein. The documents may be digitized from physical documents, such as documents that have been scanned by a scanner device. The documents may also be digitally generated according to document or word processing software. In some situations, the documents may be processed according to an optical character recognition (OCR) system or process. The documents may include data or metadata indicating at least words and lines that have been detected in the documents according to the OCR.
- (13) The documents may have indications of where the words and lines are located within the document. The words and lines may be defined by positional information that indicate respective locations in the respective documents. For example, the positional information may include (X, Y) coordinates that indicate relative positions within a document. As another example, the positional information may also include height and width values that indicate a size of a bounding box for the words and lines.

(14) The document layout system may receive the document that has been processed with OCR. In some implementations, a machine learning model may be implemented and applied to the document to generate a graph representation of the document. The graph representation may include a plurality of nodes and one or more edges, where individual edges are connected to at least two of the nodes. The nodes may represent text sections comprising one or more words or lines that collectively form a text section of the document. The graph representation may be visualized as a line graph that traces through the text sections of the document to represent a reading order or a layout of the document.

(15) Another machine learning model may be applied to determine whether particular groupings of words or lines collectively form a text section. For example, a particular group of lines may be interpreted to be a sentence or paragraph based on their collective arrangement on the document. The sentence or paragraph may be designated as a text section for the purpose of organization and layout in the document. The machine learning model may also be configured to generate a graph representation of the document. In some implementations, the graph representation may have one or more nodes and one or more edges connected to respective ones of the one or more nodes. The nodes may comprise properties of the text from the document.

(16) The machine learning model may classify text sections based on characteristics of the features of the nodes. For example, classifications of the text sections may include title, paragraph, body text, etc. The machine learning model may also determine a reading order and a layout of the document based on the classified text sections and the graph representation of the document.

(17) In one aspect, a system is described. The system may include one or more processors and a memory storing instructions that, when executed on or across the one or more processors, cause the one or more processors to perform operations for a document layout system. The document layout system is configured to receive a document comprising text. The document layout system is also configured to apply an optical character recognition (OCR) technique to identify the text in the document. The document layout system is further configured to generate a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text identified according to the OCR technique. The document layout system is also configured to apply a graph convolutional network (GCN) machine learning model to the graph representation to identify a layout of different sections of the document according to respective merge inferences determined for individual ones of the plurality of edges. The document layout system is also configured to provide the layout of different sections of the document.

(18) In another aspect, a method is described. The method may include performing, with one or more computing devices: receiving a document comprising text. The method may also include generating a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text identified according to a text detection technique. The method may further include applying a graph convolutional network (GCN) machine learning model to the graph representation to identify a hierarchy of different sections of the document according to respective merge inferences determined for individual ones of the plurality of edges. The method may also include providing the hierarchy of different sections of the document.

(19) In yet another aspect, one or more computer-readable storage media storing instructions that, when executed on or across one or more processors, cause the one or more processors to perform operations is described. The operations may include in response to receiving a document comprising text, generating a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the

text identified according to a text detection technique. The operations may also include applying a graph convolutional network (GCN) machine learning model to the graph representation to identify a hierarchy of different sections of the document according to respective merge inferences determined for individual ones of the plurality of edges. The operations may further include providing the hierarchy of different sections of the document.

(20) FIG. 1 is a block diagram of system **100** configured to determine text section categories and layouts of documents, according to some embodiments. Various aspects of system **100** may be implemented by one or more computing devices that include one or more processors and memories that store program instructions that cause the one or more processors to perform computing operations. System **100** may include document layout system **102**. In some implementations, document layout system **102** may be implemented by one or more computing devices hosted by a service provider. The service provider may include a provider network that is configured to provide access to various network-based services to various clients via a network, such as the Internet.

(21) System **100** may be configured to receive documents from clients or data stores to be processed by document layout system **102**, according to some embodiments. As an illustrative example, document **110a** may be provided to system **100** via one or more interfaces between clients and system **100**. For example, the interfaces may include application programmatic interfaces (APIs) that are exposed or made available for access to system **100**. Document **110a** may be a digitized document based on capturing a physical document. In many situations, document **110a** may be an optically captured image of the physical document. For example, document **110a** may be a scanned copy of a printed document. Thus, document **110a** may initially lack plain text data based on being a scanned copy.

(22) Document layout system **102** may also include optical character recognition (OCR) **112** configured to detect and process text that may be found in documents, such as document **110a**, according to some embodiments. OCR **112** may include one or more processes or algorithms configured to detect text elements included in document **110a**. For example, OCR **112** may include an OCR process configured to detect the text elements in document **110a**. In some implementations, OCR **112** may cause the detected text elements of document **110a** to be handled as plain text by one or more other software functions, such as text selection, copying, editing, or any other similar type of text-based functionality. OCR **112** may generate document **110b** according to the OCR process, where document **110b** has text elements that were determined by the OCR process. In some implementations, document **110b** may be stored in place of document **110a** in a data store. For example, document **110a** may be overwritten by document **110b**. In some implementations, document layout system **102** may include an object detection model configured to detect text sections based on analyzing the text in the document. The object detection model may detect the objects separately from graph representations of the documents. OCR **112** and the object detection model may be combined to improve text detection by having multiple detection models applied to the same document.

(23) OCR **112** may identify various text components of document **110b**, according to some embodiments. For example, OCR **112** may identify individual words and individual lines of document **110b**. In some embodiments, the text components may be indicated by metadata. For example, indications of the text components may be appended to the metadata of document **110b**. The indications may include representations of bounding boxes that encapsulate the individual words and the individual lines. For example, a bounding box may be defined by (X,Y) coordinates that indicate relative positions in document **110b**. As another example, the bounding box may also be defined by width and height dimensions that extend from the (X,Y) coordinates. In some implementations, the bounding box may be defined by a combination of (X, Y, W, H) coordinates that describe the (X,Y) location and the width and height of the resulting bounding box.

(24) Document **110b** may be provided to document layout system **102** for determination of any text sections and layouts contained therein, according to some embodiments. Document layout system

102 may include one or more machine learning models configured to perform machine learning-based functions on input documents, such as document **110b**. In some embodiments, document layout system **102** may include graph generation **114**, graph convolutional network (GCN) **116**, and document layout **122**.

(25) Graph generation **114** may be configured to generate graph representations of documents. For example, graph generation **114** may be configured to generate graph representation **116** based on document **110b**. Graph generation **114** may include one or more neural networks configured to determine relationships between text elements found in documents. For example, a neural network may be configured to determine connectivity between different text elements. In some implementations, graph generation **114** may establish the text elements, such as words or lines, as nodes in graph representation **116**. Graph generation **114** may determine connections between various text elements and record the connections as graph edges in graph representation **116**. In some embodiments, the connections may be determined based on relative distances between bounding boxes of the text elements. For example, the Euclidian distance between center points of two bounding boxes may serve as a basis for determining the connections. As another example, a shortest distance between two bounding boxes, such as between outer edges of the bounding boxes, may serve as a basis for determining the connections.

(26) Based on the determined connections, graph generation **114** may generate graph representation **116** of document **110b**. In some implementations, graph representation **116** may be characterized as a line graph where nodes are interconnected with graph edges, where nodes are points in Euclidean space, and graph edges are lines connecting the points. Different connectivity policies may be implemented to determine the connections. The connectivity policy may be selected by the client or document layout system **102** from a list of connectivity policy options. For example, the connectivity policies may include an all-to-all algorithm, where each of the nodes is connected to the other nodes. As another example, the connectivity policies may include a K-nearest neighbors algorithm, where a node is connected to an integer k quantity of nearest other nodes. As yet another example, the connectivity policies may include a beta skeleton algorithm, where a node is connected to another node based on relative angles between the node and multiple other nodes based on a numerical parameter f.

(27) Graph generation **114** may determine text sections of documents based on a hierarchy of words, lines and sections, according to some embodiments. The text sections may be determined based on groupings of words or lines. For example, a text section may be determined based on a subset of lines in the document being grouped together. The text sections may include paragraphs that have multiple lines of text. A reader may construe a contiguous text section as a paragraph. In some implementations, graph generation **114** may include a clustering function to determine clusters of text that may be respective text sections, such as paragraphs or text blocks. For example, the clustering function may determine that a plurality of words or lines collectively form a contiguous text section.

(28) Nodes of graph representation **116** may include multiple features that define at least one aspect of the respective text element represented by the respective node, according to some embodiments. For example, the features of a node may include a visual feature, a geometric feature, and a textual feature. The visual feature may include an image copy of the text element as it appears in document **110b**. The image copy may be obtained from the document **110b**. In some implementations, the image copy may be generated based on selectively cropping an image of document **110b** based on the bounding box of the text element. The cropped image may also be an input to a residual neural network to further implement deep learning of image data. The geometric feature may include geometric definitions of the bounding box of the text element. For example, the geometric definitions may include (X, Y, W, H) coordinates for (X,Y) positioning with a width and height of the bounding box. The textual feature may include the text element transcribed in plain text according to an OCR process, such as by OCR **112**. As another example, the textual feature may

include text tokens for the portion of the text generated by OCR **112**.

(29) Document layout system **102** may also include GCN **118**, according to some embodiments. GCN **118** may be configured to apply one or more machine learning models to graph representation **116** to generate edge merge inferences **120** based on the edges in graph representation **116**. For example, GCN **118** may generate edge merge inferences **120** based on characteristics of the edges in graph representation **116**. In some implementations, the characteristics may include relative lengths of the graphs or relative positions of the nodes.

(30) Document layout **122** may be configured to detect text sections of document **110b** according to edge merge inferences **120** generated based on graph representation **116**, according to some embodiments. Document layout **122** may be configured to identify and classify types of text sections in documents, according to some embodiments. Document layout **122** may include section classification **126** configured to identify text sections contained in documents based on edge merge inferences **120** for graph representation **116**. For example, section classification **126** may identify that a particular text section is the title of document **110b**. As another example, section classification **126** may identify that a particular text section is a paragraph of document **110b**. The determined type may be included as part of metadata of document **110b** and associated with the respective text sections.

(31) The determined classifications may be further processed to determine a layout of document **110b**, according to some embodiments. For example, document layout **122** configured to predict or determine the layout of documents based on various criteria including the detected text sections or classifications. Document layout **122** may be configured to analyze the determined classifications of the text elements in view of graph representation **116**. For example, document layout **122** may compare the classifications against other documents and respective locations of text elements and their respective classifications. In some implementations, document layout **122** may include weights based on different types of text element classifications such that different types of classifications may increase or decrease the likelihood that a given document has a particular layout and type.

(32) Document layout system **102** may output information indicating text section categories and layout **122** of document **110a**. The information indicating text section categories and layout **122** may be provided to a data storage destination or provided in response to a client request, according to various embodiments.

(33) FIG. 2 illustrates representations **200** of a document being processed by a document layout system, according to some embodiments. The document layout system may correspond to document layout system **102** of FIG. 1. The document may be provided by a client to the document layout system via an API exposed to the client from a service provider.

(34) Document **210** is an illustrative example of a document that may be provided by the client to the document layout system for processing, according to some embodiments. The particular layout and text on document **210** are for illustrative purposes and are not intended to be limiting on the types of documents that may be provided to or processed by the document layout system. Document **210** may include title **211**, heading **213a**, body text **215a**, and heading **213b**. Document **210** may also include image **217** that has caption text **218**. Document **210** may also include subheading **213b** and text columns **219a** and **219b**.

(35) An OCR process may receive document **210** for processing. The OCR process may correspond to OCR **112** of FIG. 1, according to some embodiments. The OCR process may generate document **220** based on document **210**. The OCR process may include identifying text elements contained in document **110**. The OCR process may further identify individual words included in document **210**. As an illustrative example, the OCR process may be configured to highlight or indicate words contained in document **210** with a bounding box.

(36) Document **220** may include title **221**, heading **223a**, body text **225a**, and heading **213b**. Document **220** may also include image **227** that has caption text **228**. Document **220** may also

include subheading **223b** and text columns **229a** and **229b**. An OCR process may generate bounding boxes around individual words of document **222**. The OCR process may generate bounding boxes **222** for the words of title **221**. The OCR process may generate bounding boxes **224a** for the words of heading **223a**. The OCR process may also generate bounding boxes **226a** for body text **225a**. Image **227** does not contain any text and therefore is not represented by a bounding box. The OCR process may generate bounding boxes **224b** for heading **223b**. The OCR process may also generate bounding boxes **226b** and **226c** respectively for text columns **229a** and **229b**.

(37) Document **230** may include title **231**, heading **233a**, body text **235a**, and heading **213b**. Document **230** may also include image **237** that has caption text **238**. Document **230** may also include subheading **233b** and text columns **239a** and **239b**. An OCR process may generate bounding boxes around individual lines of document **232**. The OCR process may generate bounding boxes **232** for the line of title **231**. The OCR process may generate bounding boxes **234a** for the line of heading **233a**. The OCR process may also generate bounding boxes **236a** for body text **235a**. Image **237** does not contain any text and therefore is not represented by a bounding box. The OCR process may generate bounding boxes **234b** for heading **233b**. The OCR process may also generate bounding boxes **236b** and **236c** respectively for text columns **239a** and **239b**.

(38) Document **240** may include title **241**, heading **243a**, body text **245a**, and heading **213b**. Document **240** may also include image **247** that has caption text **248**. Document **240** may also include subheading **243b** and text columns **249a** and **249b**. A document layout process may generate bounding boxes around text sections of document **242**. The document layout process may generate bounding boxes **242** for the text section of title **241**. The document layout process may generate bounding boxes **244a** for the text section of heading **243a**. The document layout process may also generate bounding boxes **246a** for body text **245a**. Image **247** does not contain any text and therefore is not represented by a bounding box. The document layout process may generate bounding boxes **244b** for heading **243b**. The document layout process may also generate bounding boxes **246b** and **246c** respectively for text columns **249a** and **249b**.

(39) FIG. 3 illustrates graph representations **300** of documents processed by one or more aspects of a document layout system, according to some embodiments. The document layout system may correspond to document layout system **102** of FIG. 1, according to some embodiments.

(40) Document **330** may include title **331**, heading **333a**, body text **335a**, and heading **313b**. Document **330** may also include image **337** that has caption text **338**. Document **330** may also include subheading **333b** and text columns **339a** and **339b**. An OCR process may generate bounding boxes around individual lines of document **332**. The OCR process may generate bounding boxes **332** for the line of title **331**. The OCR process may generate bounding boxes **334a** for the line of heading **333a**. The OCR process may also generate bounding boxes **336a** for body text **335a**. Image **337** does not contain any text and therefore is not represented by a bounding box. The OCR process may generate bounding boxes **334b** for heading **333b**. The OCR process may also generate bounding boxes **336b** and **336c** respectively for text columns **339a** and **339b**.

(41) A graph generator, such as graph generation **114**, may be configured to generate a graph representation of document **330**, such as graph representation **310**. In different implementations, the graph generator may generate the graph representation based on documents that have bounding boxes for words, such as document **220**. As an illustrative example, graph representation **310** comprises nodes and edges, where the nodes represent text elements in document **330**, and the edges represent connections or relationships between the text elements. For example, a given node of graph representation **310** may correspond a particular text element, such as bounding box **332** for title **331**. In some implementations, the nodes may include metadata indicating multiple features that define at least one aspect of the respective text element represented by the respective node, according to some embodiments. For example, the features of a node may include a visual feature, a geometric feature, and a textual feature. The visual feature may include an image copy of the text element as it appears in document **330**. In some implementations, the image copy may be generated

based on selectively cropping an image of document **330**. The cropped image may also be an input to a residual neural network to further implement deep learning of image data. The geometric feature may include geometric definitions of the bounding box of the text element. For example, the geometric definitions may include (X, Y, W, H) coordinates for (X,Y) positioning with a width and height of the bounding box. The textual feature may include the text element transcribed in plain text according to the OCR process.

(42) As an illustrative example, the node for bounding box **334b** have multiple edges directed to bounding boxes **336b** of column **339a** and bounding boxes **336c** of column **339b**. In some situations, the graph generator may attempt to infer a reading order between different text elements. A human reader would be able to determine which of text columns **339a** or **339b** are supposed to be read after heading **333b**. However, the graph generator may make inferences based on analysis of previous documents. The directionality and shape of graph representation **310** may indicate inferences for the reading order of document **330**.

(43) Document **340** may include title **341**, heading **343a**, body text **345a**, and heading **313b**. Document **340** may also include image **347** that has caption text **348**. Document **340** may also include subheading **343b** and text columns **349a** and **349b**. A document layout process may generate bounding boxes around text sections of document **342**. The document layout process may generate bounding boxes **342** for the text section of title **341**. The document layout process may generate bounding boxes **344a** for the text section of heading **343a**. The document layout process may also generate bounding boxes **346a** for body text **345a**. Image **347** does not contain any text and therefore is not represented by a bounding box. The document layout process may generate bounding boxes **344b** for heading **343b**. The document layout process may also generate bounding boxes **346b** and **346c** respectively for text columns **349a** and **349b**.

(44) Graph generation **114** may generate a graph generation of document **340**, such as graph representation **320**. As an illustrative example, graph representation **320** comprises nodes and edges, where the nodes represent text elements in document **340**, and the edges represent connections or relationships between the text elements. For example, a given node of graph representation **330** may correspond a particular text element, such as bounding box **342** for title **341**. Edge **322a** is depicted as a dashed line to indicate a connection between respective nodes corresponding to bounding boxes **342** and **344a**. Line **324a** is depicted as a solid line to indicate a reading order through body text **345a** represented by bounding box **346a**.

(45) Edge **322b** may pass through image **347** based on the OCR process bypassing non-text content found in document **340**. Based on a determination that image **347** is to be bypassed, caption text **348** may also be excluded from graph representation **320**. The node corresponding to heading **343b** represented by bounding box **344b** may have multiple edges based on subsequent bounding boxes **346b** and **346c**. As an illustrative example, the node for bounding box **344b** have multiple edges directed to bounding boxes **346b** of column **349a** and bounding boxes **346c** of column **349b**. In some situations, the graph generator may attempt to infer a reading order between different text elements. A human reader would be able to determine which of text columns **349a** or **349b** are supposed to be read after heading **343b**. However, the graph generator may make inferences based on analysis of previous documents. The directionality and shape of graph representation **320** may indicate inferences for the reading order of document **340**, where dashed lines indicate edge connections between nodes having reading orders indicated by solid lines.

(46) FIG. 4 illustrates a block diagram **400** of bounding boxes **402** and **412** that represent text elements in a document that is processed by a document layout system, according to some embodiments.

(47) Bounding box **402** may be configured to encapsulate or identify text **404** in the document. Text **404** may include an image representation of text **404** as captured from the document. Text **404** may include a plain text representation of text **404** as determined by an OCR process. Bounding box **402** may be defined by coordinates **406**. For example, coordinates **406** may include (X, Y, W, H)

coordinates for (X,Y) positioning with a width and height of bounding box **402**.

(48) Bounding box **412** may be configured to encapsulate or identify text **414** in the document. Text **414** may include an image representation of text **414** as captured from the document. Text **414** may include a plain text representation of text **414** as determined by an OCR process. Bounding box **412** may be defined by coordinates **416**. For example, coordinates **416** may include (X, Y, W, H) coordinates for (X,Y) positioning with a width and height of bounding box **412**.

(49) Bounding boxes **402** and **412** may be associated with respective nodes in a graph representation, such as graph representation **116** of FIG. 1, graph representation **310** or graph representation **320** of FIG. 3, according to some embodiments. Connection line **410** may represent a connection between bounding boxes **402** and **412**. In some implementations, connection line **410** may be determined based on a distance between respective center points of bounding boxes **402** and **412**.

(50) In other implementations, connection line **410** may be determined based on a closest distance between any portion of bounding boxes **402** and **412**. Bounding boxes **402** and **412** may have box edges that define respective perimeters. As an illustrative example, box edge **408** may represent the closest box edge of bounding box **402** to bounding box **412**, and box edge **418** may represent the closest box edge of bounding box **412** to bounding box **402**.

(51) FIG. 5 is a logical block diagram illustrating a provider network that implements different services including a code development service, according to some embodiments. A provider network **500** (which may, in some implementations, be referred to as a “cloud provider network” or simply as a “cloud”) refers to a pool of network-accessible computing resources (such as compute, storage, and networking resources, applications, and services), which may be virtualized or bare-metal. The provider network **500** can provide convenient, on-demand network access to a shared pool of configurable computing resources that can be programmatically provisioned and released in response to customer commands. These resources can be dynamically provisioned and reconfigured to adjust to variable load.

(52) The provider network **500** can be formed as a number of regions, where a region is a separate geographical area in which the cloud provider clusters data centers. Each region can include two or more availability zones connected to one another via a private high speed network, for example a fiber communication connection. An availability zone (also known as an availability domain, or simply a “zone”) refers to an isolated failure domain including one or more data center facilities with separate power, separate networking, and separate cooling from those in another availability zone. Preferably, availability zones within a region are positioned far enough away from one other that the same natural disaster should not take more than one availability zone offline at the same time. Customers can connect to availability zones of the provider network **500** via a publicly accessible network (e.g., the Internet, a cellular communication network). Regions are connected to a global network which includes private networking infrastructure (e.g., fiber connections controlled by the cloud provider) connecting each region to at least one other region. The provider network **500** may deliver content from points of presence outside of, but networked with, these regions by way of edge locations and regional edge cache servers. This compartmentalization and geographic distribution of computing hardware enables the provider network **500** to provide low-latency resource access to customers on a global scale with a high degree of fault tolerance and stability.

(53) As noted above, provider network **500** may implement various computing resources or services, such as machine learning service **510**, data storage service **530** and other service(s) **540** which may be any other type of network-based services, including various other types of compute, data processing, machine learning, analysis, communication, event handling, visualization, and security services not illustrated).

(54) In various embodiments, the components illustrated in FIG. 5 may be implemented directly within computer hardware, as instructions directly or indirectly executable by computer hardware

(e.g., a microprocessor or computer system), or using a combination of these techniques. For example, the components of FIG. 5 may be implemented by a system that includes a number of computing nodes (or simply, nodes), each of which may be similar to the computer system embodiment illustrated in FIG. 8 and described below. In various embodiments, the functionality of a given system or service component (e.g., a component of machine learning service 510) may be implemented by a particular node or may be distributed across several nodes. In some embodiments, a given node may implement the functionality of more than one service system component (e.g., more than one data store component).

(55) Machine learning service 510 may be implemented by provider network 500, according to some embodiments. Machine learning service 510 may implement various features for generating, training and deploying machine learning models for different systems, applications, or devices. Machine learning service 510 may include optical character recognition service 502, graph convolutional network 512, and document layout service 516, according to some embodiments.

(56) Optical character recognition service 502 may be configured to receive document 504 from clients 550. Optical character recognition service 502 may implement an OCR process to generate OCR document 506 based on document 504. OCR document 506 may identify text elements with bounding boxes determined according to the OCR process.

(57) Optical character recognition service 502 may provide OCR document 506 to graph convolutional network 512. Graph convolutional network 512 may generate graph 514 of OCR document 506. Graph 514 may include nodes that represent text elements of OCR document 506 and edges that connect the text elements.

(58) Document layout service 516 may be configured to determine document text classification(s) 518 based on graph 514 provided by graph convolutional network 514, according to some embodiments. For example, document layout service 516 may determine document text classification(s) 518 based on the nodes and edges indicated by graph 514 to determine that a particular text element has a particular text classification. Document text classification(s) 518 may be returned to client 550 or stored to data storage service 530 in addition to OCR document 506.

(59) Generally speaking, clients 550 may encompass any type of client configurable to submit network-based requests to provider network 500 via network 560, including requests for services (e.g., a request for code search or suggestion, etc.). For example, a given client 550 may include a suitable version of a web browser, or may include a plug-in module or other type of code module that may execute as an extension to or within an execution environment provided by a web browser. Alternatively, a client 550 may encompass an application (or user interface thereof), a media application, an office application or any other application that may make use of resources in provider network 500 to implement various applications. In some embodiments, such an application may include sufficient protocol support (e.g., for a suitable version of Hypertext Transfer Protocol (HTTP)) for generating and processing network-based services requests without necessarily implementing full browser support for all types of network-based data. That is, client 550 may be an application may interact directly with provider network 500. In some embodiments, client 550 may generate network-based services requests according to a Representational State Transfer (REST)-style network-based services architecture, a document- or message-based network-based services architecture, or another suitable network-based services architecture.

(60) In some embodiments, a client 550 may provide access to provider network 500 to other applications in a manner that is transparent to those applications. For example, client 550 may integrate with machine learning service 510. However, the operating system or file system may present a different storage interface to applications, such as a conventional file system hierarchy of files, directories and/or folders. In such an embodiment, applications may not need to be modified to make use of the storage system service model. Instead, the details of interfacing to the data storage service may be coordinated by client 550 and the operating system or file system on behalf of applications executing within the operating system environment.

(61) Clients **550** may convey network-based services requests to and receive responses from provider network **500** via network **560**. In various embodiments, network **560** may encompass any suitable combination of networking hardware and protocols necessary to establish network-based communications between clients **550** and provider network **500**. For example, network **560** may generally encompass the various telecommunications networks and service providers that collectively implement the Internet. Network **560** may also include private networks such as local area networks (LANs) or wide area networks (WANs) as well as public or private wireless networks. For example, both a given client **550** and provider network **500** may be respectively provisioned within enterprises having their own internal networks. In such an embodiment, network **560** may include the hardware (e.g., modems, routers, switches, load balancers, proxy servers, etc.) and software (e.g., protocol stacks, accounting software, firewall/security software, etc.) necessary to establish a networking link between given client **550** and the Internet as well as between the Internet and provider network **500**. It is noted that in some embodiments, clients **550** may communicate with provider network **500** using a private network rather than the public Internet.

(62) In some embodiments, provider network **500** may include the hardware (e.g., modems, routers, switches, load balancers, proxy servers, etc.) and software (e.g., protocol stacks, accounting software, firewall/security software, etc.) necessary to establish a networking links between different components of provider network **500**, such as virtualization hosts, control plane components as well as external networks **560** (e.g., the Internet). In some embodiments, provider network **500** may employ an Internet Protocol (IP) tunneling technology to provide an overlay network via which encapsulated packets may be passed through the internal network using tunnels. The IP tunneling technology may provide a mapping and encapsulating system for creating an overlay network and may provide a separate namespace for the overlay layer and the internal network layer. Packets in the overlay layer may be checked against a mapping directory to determine what their tunnel target should be. The IP tunneling technology provides a virtual network topology; the interfaces that are presented to clients **550** may be attached to the overlay network so that when a client **550** provides an IP address that they want to send packets to, the IP address is run in virtual space by communicating with a mapping service that knows where the IP overlay addresses are.

(63) FIG. **6** illustrates an interface **600** for a document layout service, according to some embodiments. The document layout service may correspond to document layout system **102** of FIG. **1** or machine learning service **510** of FIG. **5**, according to some embodiments. Interface **600** may be presented via one or more client-facing interfaces, such as a web interface or an API.

(64) Interface **600** may include a file upload element **602**, according to some embodiments. File upload element **602** may include a file selection element **604** configured to receive input from the client. The client may provide a document via the file selection element **604** to be uploaded to the document layout service for processing. The file upload element **602** may include an upload button **606**. In response to a user interaction with the upload button **606**, the interface **600** may obtain the document and provide the document to the document layout service.

(65) After processing the document, the document layout service may provide document analysis **610** to the client via interface **600**, according to some embodiments. Document analysis **610** may include a display element for document preview **612**. In some implementations, document preview **612** may include a thumbnail representation of the document. The thumbnail representation may include bounding boxes around words, lines, or text sections based on a requested granularity. Document analysis **610** may indicate predicted classification(s) **614** for the document. Predicted classification(s) **614** may be based on a graph convolutional network and/or a document layout service configured to analyze the document. Predicted classification(s) **614** may include classifications **616a**, **616b**, **616c**, and any other number of classifications. Predicted classification(s) **614** may also indicate a likelihood of each of classifications **616a**, **616b**, and **616c**.

expressed as a percentage or ranking that a given classification is more likely to be correct.

(66) FIG. 7 is a block diagram of a method **700** for determining a document layout based on text elements contained within the document, according to some embodiments. The method **700** may be implemented by one or more computing devices. The method **700** may be implemented by document layout system **102** of FIG. 1 or machine learning service **510** of FIG. 5, according to some embodiments.

(67) The method **700** may include receiving, at a document layout system, a document comprising text, at **702**. The method **700** may also include applying an optical character recognition (OCR) technique to identify the text elements in the document, at **704**. The method may further include generating a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text identified according to the OCR technique, at **706**.

(68) The method **700** may also include applying a graph cluster network (GCN) machine learning model to the graph representation to identify a layout of different sections of the document according to respective merge inferences determined for individual ones of the plurality of edges. The method **700** may further include providing the layout of different sections of the document, at **710**.

(69) FIG. 8 shows an example computing device to implement the various techniques described herein, according to some embodiments. For example, in one embodiment, the algorithm execution management system described above may be implemented by a computer device, for instance, a computer device as in FIG. 8 that includes one or more processors executing program instructions stored on a computer-readable storage medium coupled to the processors. In the illustrated embodiment, computer system **800** includes one or more processors **810** coupled to a system memory **820** via an input/output (I/O) interface **830**. Computer system **800** further includes a network interface **840** coupled to I/O interface **830**. While FIG. 8 shows computer system **800** as a single computing device, in various embodiments a computer system **800** may include one computing device or any number of computing devices configured to work together as a single computer system **800**.

(70) In various embodiments, computer system **800** may be a uniprocessor system including one processor **810**, or a multiprocessor system including several processors **810** (e.g., two, four, eight, or another suitable number). Processors **810** may be any suitable processors capable of executing instructions. For example, in various embodiments, processors **810** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors **810** may commonly, but not necessarily, implement the same ISA.

(71) System memory **820** may be one embodiment of a computer-accessible medium configured to store instructions and data accessible by processor(s) **810**. In various embodiments, system memory **820** may be implemented using any non-transitory storage media or memory media, such as magnetic or optical media, e.g., disk or DVD/CD coupled to computer system **800** via I/O interface **830**. A non-transitory computer-accessible storage medium may also include any volatile or non-volatile media such as RAM (e.g. SDRAM, DDR SDRAM, RDRAM, SRAM, etc.), ROM, etc., that may be included in some embodiments of computer system **800** as system memory **820** or another type of memory. Further, a computer-accessible medium may include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **840**. In the illustrated embodiment, program instructions (e.g., code) and data implementing one or more desired functions, such as the algorithm execution management system described above in FIGS. 1-6, are shown stored within system memory **830** as code **825** and data **826**.

(72) In one embodiment, I/O interface **830** may be configured to coordinate I/O traffic between processor **810**, system memory **820**, and any peripheral devices in the device, including network interface **840** or other peripheral interfaces. In some embodiments, I/O interface **830** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **820**) into a format suitable for use by another component (e.g., processor **810**). In some embodiments, I/O interface **830** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **830** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **830**, such as an interface to system memory **820**, may be incorporated directly into processor **810**.

(73) Network interface **840** may be configured to allow data to be exchanged between computer system **800** and other devices **860** attached to a network or networks **850**. In various embodiments, network interface **840** may support communication via any suitable wired or wireless general data networks, such as types of Ethernet network, for example. Additionally, network interface **840** may support communication via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks, via storage area networks such as Fiber Channel SANs, or via any other suitable type of network and/or protocol.

(74) In some embodiments, system memory **820** may be one embodiment of a computer-accessible medium configured to store program instructions and data as described above for FIG. **1-8**.

Generally speaking, a computer-accessible medium may include non-transitory storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD coupled to computer system **800** via I/O interface **830**. A non-transitory computer-accessible storage medium may also include any volatile or non-volatile media such as RAM (e.g. SDRAM, DDR SDRAM, RDRAM, SRAM, etc.), ROM, etc., that may be included in some embodiments of computer system **800** as system memory **820** or another type of memory. Further, a computer-accessible medium may include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **840**.

(75) Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g. SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc., as well as transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

(76) The various systems and methods as illustrated in the figures and described herein represent example embodiments of methods. The systems and methods may be implemented manually, in software, in hardware, or in a combination thereof. The order of any method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

(77) Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. It is intended that the embodiments embrace all such modifications and changes and, accordingly, the above description to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A system, comprising: one or more processors; and a memory storing instructions that, when executed on or across the one or more processors, cause the one or more processors to: receive a

document comprising text; apply an optical character recognition (OCR) technique to identify the text in the document; generate a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text identified according to the OCR technique; apply a graph convolutional network (GCN) machine learning model to the graph representation to identify a layout of different sections of the document according to respective merge inferences generated by the GCN for individual ones of the plurality of edges; and provide the layout of different sections of the document.

2. The system of claim 1, wherein, to apply the GCN machine learning model, the memory further comprises instructions that, when executed on or across the one or more processors, cause the one or more processors to: identify individual ones of the different sections based on a determination that a respective subset of the plurality of nodes is to be grouped together.

3. The system of claim 2, wherein the memory further comprises instructions that, when executed on or across the one or more processors, cause the one or more processors to: identify respective section classifications for the different sections of the document based on characteristics of the respective nodes for the different sections and the corresponding, respective edges connected to the respective nodes; and determine the layout for the document based on the respective section classifications.

4. The system of claim 1, wherein individual ones of the plurality of nodes comprise: respective image data obtained from the document for the portion of the text; a respective bounding box for the portion of the text; and a respective one or more text tokens for the portion of the text generated by the OCR technique.

5. The system of claim 1, wherein to generate the graph representation the memory further comprises instructions that, when executed on or across the one or more processors, cause the one or more processors to: for individual nodes of the plurality of nodes: determine respective edges to connect to one or more other nodes of the plurality of nodes according to a connectivity policy selected from a plurality of connectivity policy options.

6. A method, comprising: performing, with one or more computing devices: receiving a document comprising text; generating a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text identified according to a text detection technique; applying a graph convolutional network (GCN) machine learning model to the graph representation to identify a hierarchy of different sections of the document according to respective merge inferences generated by the GCN for individual ones of the plurality of edges; and providing the hierarchy of different sections of the document.

7. The method of claim 6, wherein applying the GCN machine learning model comprises: identifying individual ones of the different sections based on a determination that a respective subset of the plurality of nodes is to be grouped together.

8. The method of claim 6, wherein applying the GCN machine learning model comprises: identifying respective section classifications for the different sections of the document based on characteristics of the respective nodes for the different sections and the corresponding, respective edges connected to the respective nodes; and determining the hierarchy for the document based on the respective section classifications.

9. The method of claim 6, wherein individual ones of the plurality of nodes comprise: respective image data obtained from the document for the portion of the text; a respective bounding box for the portion of the text; and a respective one or more text tokens for the portion of the text generated by the OCR technique.

10. The method of claim 6, wherein generating the graph representation comprises: for individual nodes of the plurality of nodes: determining respective edges to connect to one or more other nodes

of the plurality of nodes according to a connectivity policy selected from a plurality of connectivity policy options.

11. The method of claim 6, further comprising: applying an optical character recognition (OCR) technique to identify the text and the different sections in the document in response to receiving the document from a client; and generating respective bounding boxes for respective portions of the text.

12. The method of claim 6, further comprising: determining a reading order of the document based at least in part on the hierarchy of the different sections; and providing information indicating the reading order of the document.

13. The method of claim 6, further comprising: applying an object detection model to the text in the document to identify the different sections in the document based on the image of the document.

14. One or more non-transitory computer-readable storage media storing instructions that, when executed on or across one or more processors, cause the one or more processors to: in response to receiving a document comprising text, generate a graph representation of the document, wherein the graph representation comprises a plurality of nodes and a plurality of edges that connect different ones of the plurality of nodes, wherein individual ones of the nodes correspond to different portions of the text identified according to a text detection technique; apply a graph cluster network (GCN) machine learning model to the graph representation to identify a hierarchy of different sections of the document according to respective merge inferences generated by the GCN for individual ones of the plurality of edges; and provide the hierarchy of different sections of the document.

15. The one or more non-transitory computer-readable storage media of claim 14, wherein to apply the GCN machine learning model, the one or more computer-readable storage media further comprise instructions that, when executed on or across the one or more processors, cause the one or more processors to: identify individual ones of the different sections based on a determination that a respective subset of the plurality of nodes is to be grouped together.

16. The one or more non-transitory computer-readable storage media of claim 14, wherein to apply the GCN machine learning model, the one or more computer-readable storage media further comprise instructions that, when executed on or across the one or more processors, cause the one or more processors to: identify respective section classifications for the different sections of the document based on characteristics of the respective nodes for the different sections and the corresponding, respective edges connected to the respective nodes; and determine the hierarchy for the document based on the respective section classifications.

17. The one or more non-transitory computer-readable storage media of claim 14, wherein individual ones of the plurality of nodes comprise: respective image data cropped from the document for the portion of the text; a respective bounding box for the portion of the text; and a respective one or more text tokens for the portion of the text generated by the OCR technique.

18. The one or more non-transitory computer-readable storage media of claim 14, wherein to generate the graph representation, the one or more computer-readable storage media further comprise instructions that, when executed on or across the one or more processors, cause the one or more processors to: for individual nodes of the plurality of nodes: determine respective edges to connect to one or more other nodes of the plurality of nodes according to a connectivity policy selected from a plurality of connectivity policies.

19. The one or more non-transitory computer-readable storage media of claim 14, further comprising instructions that, when executed on or across the one or more processors, cause the one or more processors to: apply an optical character recognition (OCR) technique to identify the text and the different sections in the document in response to receiving the document from a client; and generate respective bounding boxes for respective portions of the text.

20. The one or more non-transitory computer-readable storage media of claim 14, further comprising instructions that, when executed on or across the one or more processors, cause the one

or more processors to: determine a reading order of the document based at least in part on the hierarchy of the different sections; and provide information indicating the reading order of the document.
